

# Federico Melo Barrero

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## EXPERIENCE

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### Canals AI

Dec 2024 – Present

Coral Gables, FL, USA · Remote · Full-time

*Software Engineer II* · Parsing

Jul 2025 – Present

- Owned parsing services for the live voice and inquiries agent systems, as part of a pipeline that processed 20000+ orders daily across 115+ wholesale distributors and 9000+ sales reps
- Built the parsing and review layers of a real-time voice service (WebSocket orchestration, speech-to-text, speaker diarization, concurrency control). Extracted ~21 products per call from live phone calls with 75%+ retained through rep review
- Integrated tools (web search, spec sheet retrieval, ERP lookup, order tracking) into an inquiries AI agent that addressed customer questions on orders, materials, and logistics. Achieved 97% of answers citing a source and 80% rated positive by reps
- Contributed to Canals' shared agent framework (Mastra), adding extended thinking, reasoning effort, image/PDF input, temperature, and tool concurrency. Migrated four Parsing agents off hand-rolled, unbounded loops onto bounded runs with step and token ceilings
- Co-built an autonomous agent (Claude Agent SDK) that polled tickets, debugged parsing failures, and shipped prompt-level fixes for review, re-parsing each order to verify its own fix
- One of 4 engineers on the 10-person Parsing team (~60 engineers total) holding merge approval rights on the main branch.

*Software Engineer I* · Parsing

Dec 2024 – Jul 2025

- Maintained a parsing pipeline supporting 19 mimetypes with intricate formats (e.g., handwritten, struck-through text, blurry and fragmented photos, cross-references to other files) in a pipeline serving ~60 wholesale distributors processing 4,000+ orders daily
- Abstracted OCR pipeline from single provider to support multiple providers, decoupling the implementation from Textract and enabling format-specific OCR selection, with a cache layer serving ~60% of requests in 0.07s versus 3.6–18.7s uncached
- Led large-scale prompt refactors improving external QA pass rate from 85% to 89%. Involved: model selection, prompt engineering, cache handling, fine-tuning, and regression testing
- Built visual highlighting systems using DOM traversal, coordinate mapping, and string comparison to map parsed products to source documents. Reduced weekly highlighting errors by 2/3
- Joined as the company's 14th engineer.

### Universidad de los Andes

Aug 2025 – Present

*Adjunct Professor* · Algorithm Design

Fall 2026

- Principal instructor for upper-division algorithm design course, delivering lectures and designing exams
- Topics: divide and conquer, dynamic programming, graph algorithms, greedy methods, NP-completeness, approximation and randomized algorithms

*Adjunct Professor* · Introduction to Programming

Fall 2025, Spring 2026

- Principal instructor for introductory undergraduate course in procedural programming using Python
- Topics: programming fundamentals and data-processing libraries Pandas and Matplotlib

### Universidad de los Andes

Bogotá, Colombia · On-site · Part-time

*Full-Stack Developer*

Jan 2024 – Dec 2024

- Designed and built full-stack student data analytics system serving 15,000+ students and 600+ faculty, replacing an external platform and saving ~\$100K USD annually
- Built parallelized data processing pipeline (Python, Polars) with complex filtering and aggregations, exposed through a unified REST API (FastAPI)
- Developed dynamic graphing, advanced filtering, report generation, and role-based dashboards in a React/TypeScript web app
- Built early-warning system for at-risk students (academic performance, dropout indicators), alerting counselors on 1,000+ critical cases
- Architected secure cross-VM communication using air-gapping and network isolation techniques to ensure compliance with student data protection requirements

Caseware International Inc.

Toronto, ON, Canada · Remote · Internship

Software Developer I

Jan 2024 – Jul 2024

- Awarded June Team Award for impactful contributions as part of Data Analytics team
- Developed Jupyter notebooks for financial and auditing data analysis
- Optimized microservices (Java, TS) performance using AWS S3 and DynamoDB
- Authored technical reports influencing team architecture decisions and best practices

Universidad de los Andes

Bogotá, Colombia · Hybrid · Part-time

Backend Developer

Aug 2023 – Dec 2023

- Designed and implemented backend architecture for the Professors' Portfolio system (Java, Spring Boot), a web application for managing faculty academic and administrative information.
- Built persistence, logic, and control layers with CRUD operations, custom logic, and RESTful APIs
- Devised efficient data migration methods for unstructured faculty data
- Collaborated on system architecture documentation and project planning roadmaps

EDUCATION

Universidad de los Andes

Bogotá, Colombia

B.Sc. Systems and Computing Engineering, Summa Cum Laude

2020 – 2024

- Cumulative GPA: 4.92 / 5.00 — Highest GPA in major in 40 years; highest in Engineering in 16 years

AWARDS & HONORS

Dejar Huella Award

Universidad de los Andes, 2026

Unianandes internal impact award, for a peer-support platform serving 56,000+ students

Award for Outstanding Saber Pro and TyT Scores 2024

Colombian Ministry of National Education, 2025

Colombia's mandatory national exit exam for graduating university students

Summa Cum Laude

Universidad de los Andes, 2025

Highest undergraduate honor; top 1% cumulative GPA among graduates of the faculty

Athletic Career Distinction

Universidad de los Andes, 2025

Awarded to one graduating student university-wide for leadership and excellence in varsity athletics

Ramón de Zubiría Award (2 times)

Universidad de los Andes, 2023, 2024

Awarded annually to the top cumulative-GPA student in the major

Semester Excellence Award (3 times)

Universidad de los Andes, Spring 2021, Fall 2021, Fall 2023

Awarded each semester to the top-GPA student in the major

TECHNICAL EXPERIENCE

AI Engineering & LLM Orchestration: Anthropic / OpenAI / Gemini / Groq APIs, Claude Agent SDK, Mastra, MCP

Backend & Systems Architecture: Python, Java, TypeScript, FastAPI, Spring Boot, WebSocket Orchestration, Concurrency Control

Data Engineering & Infrastructure: Polars, PostgreSQL, Redis, AWS (RDS, ECS, S3, DynamoDB), Docker, GitHub CI/CD

Product & Interface: React, Next.js, Svelte, Plotly, ShadCN / MUI